

Question Number : 72 Question Id : 6406531250325 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

The current share price of HCL is Rs 600. HCL is expected to pay a dividend of Rs 25 per share next year. The appropriate rate of return is 8% per year and the HCL dividend is expected to grow at a rate of 3% every year. Then, according to Gordon Growth model

Options :

6406534213118. ✖ HCL stock is underpriced.

6406534213119. ✔ HCL stock is overpriced.

6406534213120. ✖ An investor should buy HCL shares.

6406534213121. ✔ An investor should sell HCL shares.

Question Number : 73 Question Id : 6406531250326 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a put option and a call option with same strike price and time to maturity. Which of the following statements is correct?

Options :

6406534213122. ✖ If the put option is in the money, the call option will be in the money.

6406534213123. ✔ If the call option is in the money, the put option will be out of the money.

6406534213124. ✔ If the put option is at the money, the call option will also be at the money.

6406534213125. ✖ If the call option is at the money, the put option will be out of the money.

AI

Section Id :	64065387580
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	9
Number of Questions to be attempted :	9
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0

Sub-Section Number : 1
Sub-Section Id : 640653194131
Question Shuffling Allowed : No

Question Number : 74 Question Id : 6406531250327 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534213126. ✓ YES

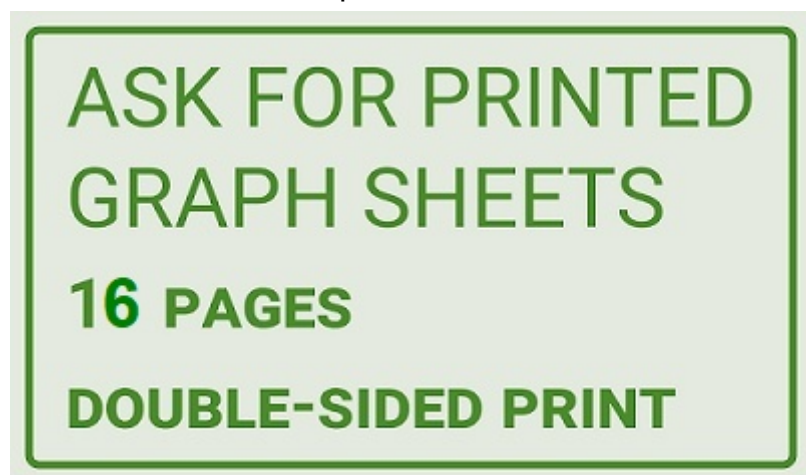
6406534213127. ✗ NO

Sub-Section Number : 2
Sub-Section Id : 640653194132
Question Shuffling Allowed : No

Question Number : 75 Question Id : 6406531250328 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question



NOTE: Each exam center must provide graph-sheets (one booklet) for each student registered to take the exam for this course. And additional blank sheets may be provided based on the student's need.

Options :

6406534213128. ✓ Printed graph sheets were provided on time.

6406534213129. ✗ Printed graph sheets were provided late.

6406534213130. ✖ Printed graph sheets were not provided.

6406534213131. ✔ I used the graph sheets.

6406534213132. ✖ I did not use graph sheets.

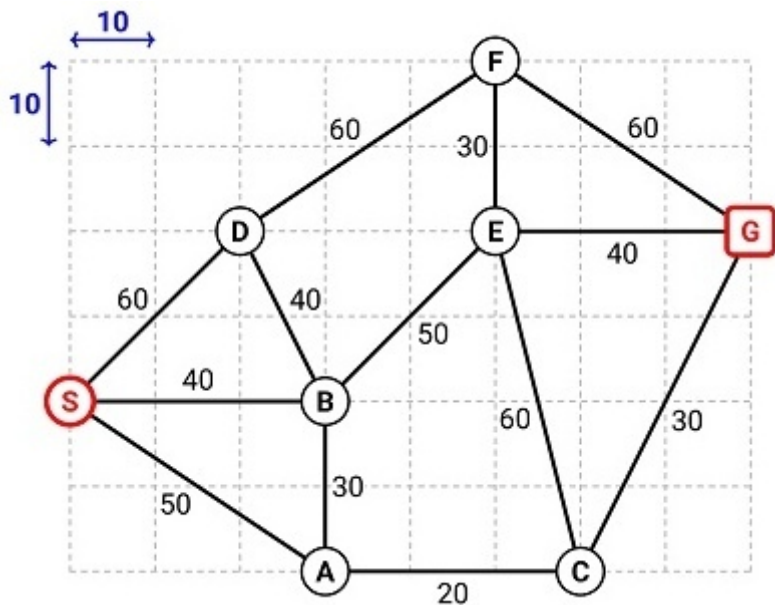
Sub-Section Number : 3
Sub-Section Id : 640653194133
Question Shuffling Allowed : No

Question Id : 6406531250329 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (76 to 80)

Question Label : Comprehension

SEARCH

The figure shows a map on a uniform grid where each tile is 10x10 in size.
The start node is S and the goal node is G.
The MoveGen function returns nodes in alphabetical order.
Use Manhattan Distance as the heuristic function.
Tie-breaker: If several nodes have the same cost, use node labels to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 76 Question Id : 6406531250330 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by the Best First Search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,D,F,G

Question Number : 77 Question Id : 6406531250331 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by A* search algorithm? Enter the path as a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,E,G

Question Number : 78 Question Id : 6406531250332 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Branch-and-Bound search algorithm? Enter the path as a comma separated list of node labels.

Use the Branch-and-Bound variation that avoids cyclic expansions like S,A,S,A,S,A,...

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,A,C,G

Question Number : 79 Question Id : 6406531250333 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given map, which algorithm finds the shortest path from S to G?

Options :

6406534213136. ✖ A* Search Algorithm

6406534213137. ✔ Branch-and-Bound Search Algorithm

6406534213138. ✖ None of these

Question Number : 80 Question Id : 6406531250334 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Select the correct statement about the given graph.

Options :

6406534213139. ✖ Heuristic is admissible.

6406534213140. ✔ Heuristic is not admissible.

6406534213141. ✖ Heuristic is admissible in some cases and not admissible in other cases.

6406534213142. ✖ There is not enough information to determine admissibility.

Sub-Section Number :

4

Sub-Section Id :

640653194134

Question Shuffling Allowed :

No

Question Id : 6406531250335 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (81 to 84)

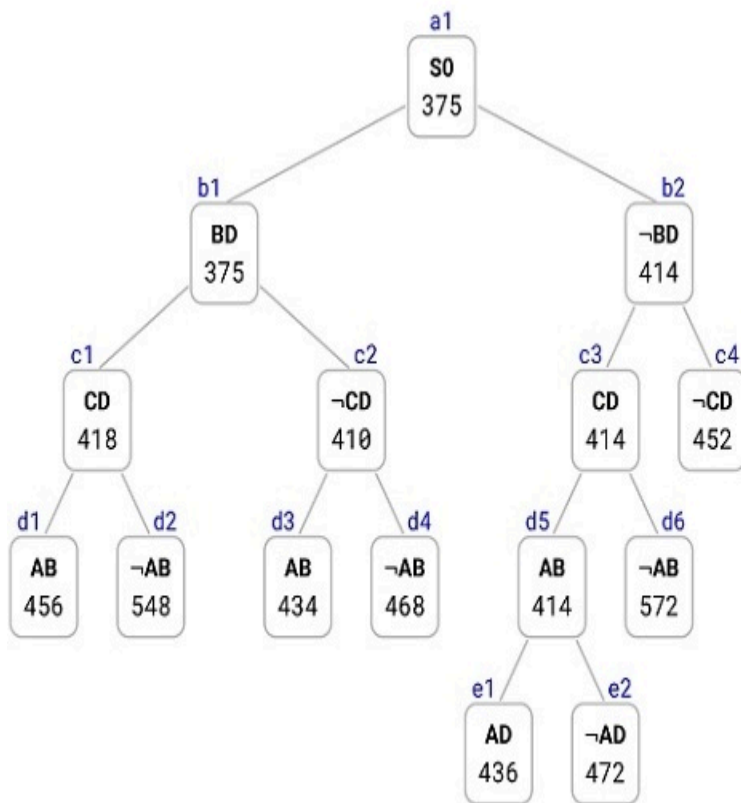
Question Label : Comprehension

TSP Branch-and-Bound

The TSP Branch-and-Bound algorithm is solving a TSP instance where the cities are A, B, C, and so on. The Branch-and-Bound search tree at the time when the algorithm has discovered the optimal tour is shown below.

Each node in the search tree displays an edge (either XY or ~XY), a cost value, and a unique reference number (a1, b1, b2, ..., c1, ..., d1, ..., e1, e2). Use the reference numbers to break ties. When required, enter the reference numbers in short answers.

What information can you glean from the search tree? Answer the sub-questions based on the information gleaned from the search tree.



Sub questions

Question Number : 81 Question Id : 6406531250336 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

In the given search problem, how many nodes were refined in order to discover the optimal tour?

Enter a number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Text Areas : PlainText

Possible Answers :

7

8

Question Number : 82 Question Id : 6406531250337 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Which node represents the optimal tour and what is the cost of the optimal tour? Enter the node reference number and the tour cost in the text box, or enter NIL if it is not possible to determine the optimal tour.

Enter a node reference number followed by tour cost, separated by comma.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: a9,42

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

d3,434

Question Number : 83 Question Id : 6406531250338 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the number of cities in the TSP instance. Enter the number of cities in the text box, or enter NIL if it is not possible to determine the number of cities.

Enter an integer.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Question Number : 84 Question Id : 6406531250339 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Start from city A, what is the path representation of the optimal tour? Enter the path

representation in the text box, or enter NIL if it is not possible to determine the optimal tour.

Enter a comma separated list of cities (city labels).
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: A,B,C

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,D,E,C

A,C,E,D,B

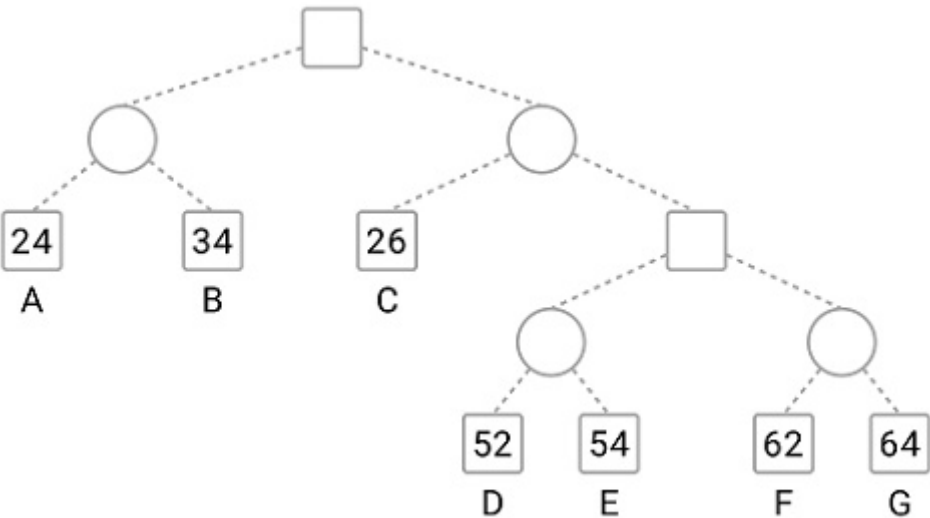
Question Id : 6406531250340 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix **Question Numbers :** (85 to 88)

Question Label : Comprehension

GAMES

The figure shows a game tree with evaluation function values at the horizon nodes.
The horizon nodes are labeled from A to G.
Use these labels to enter a horizon node or a list of horizon nodes in short answers (textbox).

Tie-breaker: when several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 85 **Question Id :** 6406531250341 **Question Type :** MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is a strategy for the MAX player?

Options :

6406534213147. ✓ C,D,E

6406534213148. ✗ A,C

6406534213149. ✓ A,B

6406534213150. ✗ A,D,F

Question Number : 86 Question Id : 6406531250342 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes in the best strategy for MAX. Enter the node labels in alphabetical order.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

C,F,G

Question Number : 87 Question Id : 6406531250343 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes pruned by Alpha-Beta.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,G

Question Number : 88 Question Id : 6406531250344 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the horizon nodes SOLVED by SSS*.

Enter a comma separated list of node labels in alphabetical order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,C,D,E

Sub-Section Number : 5

Sub-Section Id : 640653194135

Question Shuffling Allowed : No

Question Id : 6406531250345 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (89 to 91)

Question Label : Comprehension

PROBLEM DECOMPOSITION

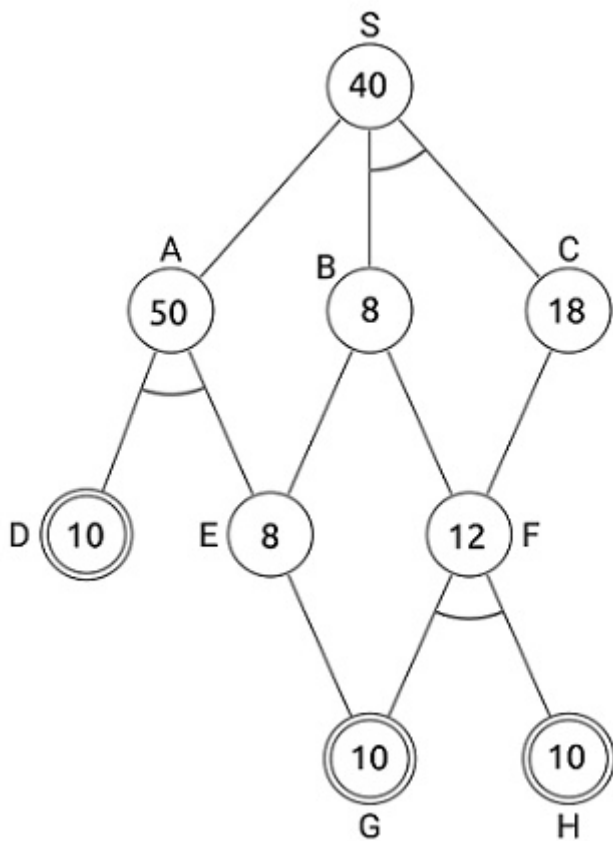
The figure shows an AND-OR graph that depicts how a problem S can be decomposed into one or more smaller problems. Nodes are uniquely identified by labels (S, A, B, ...). The number in each node is the heuristic estimate of the cost of solving that node.

Nodes shown in double lines are primitive nodes and their values are actual costs. Observe that a primitive node is added to the graph by its parent when the parent is expanded, and the primitive node is labeled as SOLVED and it will not be expanded subsequently.

The cost of each edge is 2 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.

Tie-breaker 2: For AND nodes, expand the unsolved branch with the highest cost.



Use AO* algorithm to solve S, then answer the subquestions.

Sub questions

Question Number : 89 Question Id : 6406531250346 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

List the first three nodes (including S) expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,C,F

C,F,B

Question Number : 90 Question Id : 6406531250347 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Determine the value of the start node S after each node is expanded. What are the values of S after the 1st, 2nd and 3rd nodes are expanded, respectively? Enter the 3 values in the textbox.

Enter a comma separated list of numbers.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

30,26,38

26,38,40

Question Number : 91 Question Id : 6406531250348 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the final value of the start node S computed by AO*?

Enter a number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

44

Question Id : 6406531250349 Question Type : COMPREHENSION Sub Question Shuffling

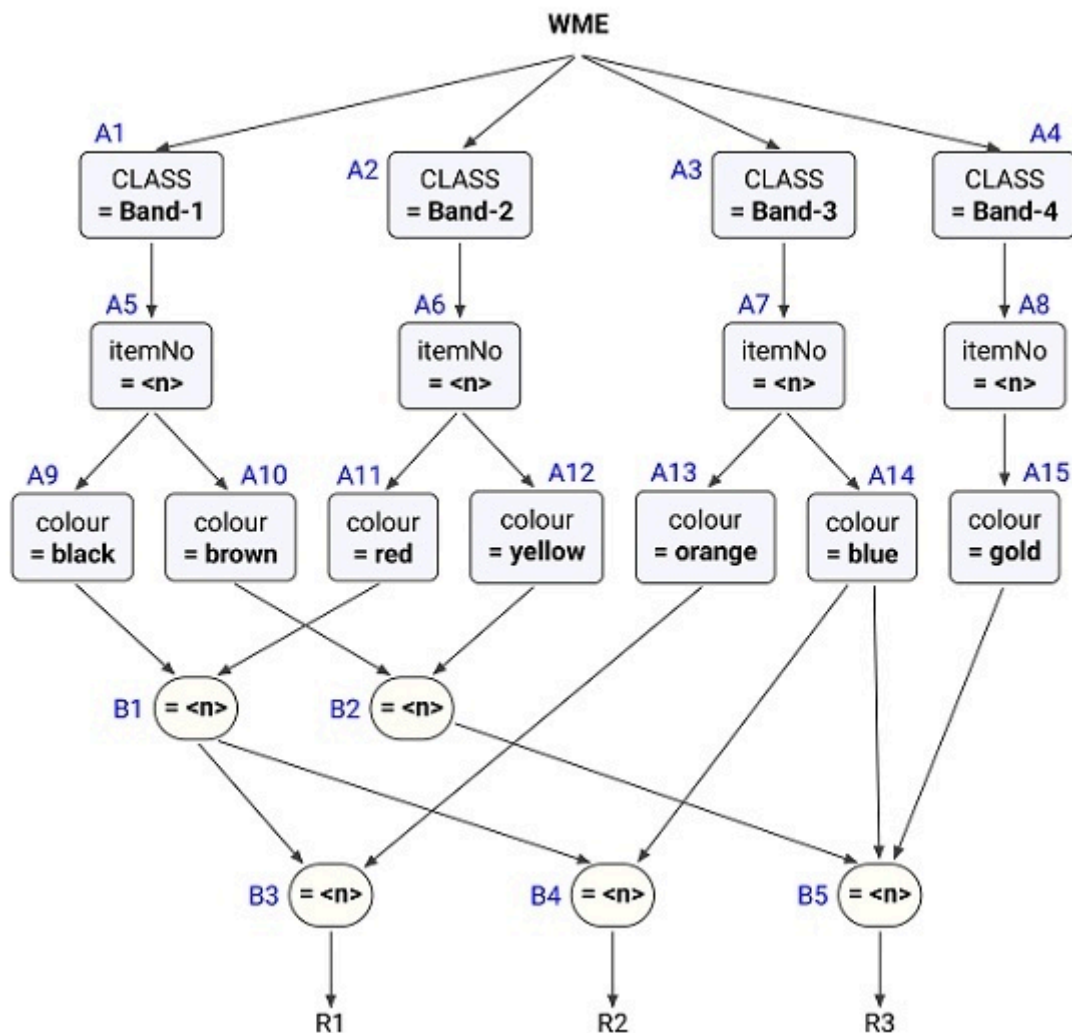
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (92 to 94)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A small part of the Rete Net for classifying resistors is shown in the figure. The labels A1, A2, ..., A10, A11, ..., B1, ..., B5 uniquely identify the nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.



Run the Rete algorithm for the Working Memory shown below, the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

201. (Band-1 ^itemNo 2B ^colour brown)
202. (Band-2 ^itemNo 2B ^colour yellow)
203. (Band-1 ^itemNo 3C ^colour black)
204. (Band-2 ^itemNo 3C ^colour yellow)
205. (Band-3 ^itemNo 3C ^colour orange)
206. (Band-3 ^itemNo 2B ^colour blue)
207. (Band-4 ^itemNo 2B ^colour gold)
208. (Band-3 ^itemNo 1A ^colour blue)
209. (Band-2 ^itemNo 1A ^colour red)
210. (Band-1 ^itemNo 1A ^colour black)

For each WME identify its location (node label) in the Rete Net, and prepare the conflict set for the first cycle, then answer the subquestions.

Sub questions

Question Number : 92 Question Id : 6406531250350 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples are in the conflict-set?

Options :

6406534213157. ✖ R1, 203, 205, 209

6406534213158. ✔ R2, 208, 209, 210

6406534213159. ✖ R3, 201, 204, 208, 207

6406534213160. ✔ R3, 201, 202, 206, 207

Question Number : 93 Question Id : 6406531250351 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Specificity** as the conflict resolution strategy then identify the rule-data tuple that will be ready to fire.

Options :

6406534213161. ✖ R1, 203, 205, 209

6406534213162. ✖ R2, 208, 209, 210

6406534213163. ✖ R3, 201, 204, 208, 207

6406534213164. ✔ R3, 201, 202, 206, 207

Question Number : 94 Question Id : 6406531250352 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** as the conflict resolution strategy then identify the rule-data tuples that will be ready to fire. If multiple rule-data tuples qualify then choose one.

Options :

6406534213165. ✖ R1, 203, 205, 209

6406534213166. ✔ R2, 208, 209, 210

6406534213167. ✖ R3, 201, 204, 208, 207

6406534213168. ✖ R3,201,202,206,207

Sub-Section Number :	6
Sub-Section Id :	640653194136
Question Shuffling Allowed :	No

Question Id : 6406531250353 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (95 to 98)
Question Label : Comprehension
AUTOMATED PLANNING

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

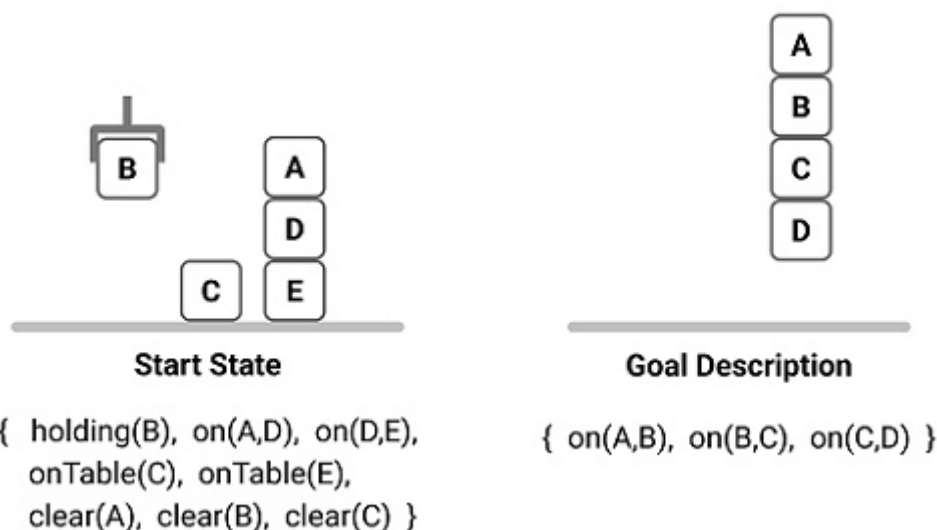
Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

Consider the planning problem with the following start state and goal description.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 95 Question Id : 6406531250354 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions in the start state?

Options :

6406534213169. ✓ Putdown(B)

6406534213170. ✓ Stack(B, C)

6406534213171. ✗ Stack(C, D)

6406534213172. ✓ Stack(B, A)

6406534213173. ✗ Stack(A, B)

Question Number : 96 Question Id : 6406531250355 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions in the goal state?

Options :

6406534213174. ✗ Putdown(B)

6406534213175. ✓ Stack(B, C)

6406534213176. ✓ Stack(C, D)

6406534213177. ✗ Stack(B, A)

6406534213178. ✓ Stack(A, B)

Question Number : 97 Question Id : 6406531250356 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex action pairs in Layer 1?

Options :

6406534213179. ✓ Stack(B,C) , Putdown(B)

6406534213180. ✓ Stack(B,A) , Putdown(B)

6406534213181. ✓ Stack(B,C) , Stack(B,A)

6406534213182. ✗ Pickup(C) , Putdown(B)

6406534213183. ✗ Unstack(A,D) , Putdown(B)

Question Number : 98 Question Id : 6406531250357 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex proposition pairs in Layer 1?

Options :

6406534213184. ✗ clear(A) , holding(B)

6406534213185. ✗ onTable(A) , on(B,C)

6406534213186. ✓ onTable(B) , on(B,C)

6406534213187. ✓ on(B,C) , on(B,A)

Sub-Section Number :

7

Sub-Section Id :

640653194137

Question Shuffling Allowed :

No

Question Id : 6406531250358 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (99 to 100)

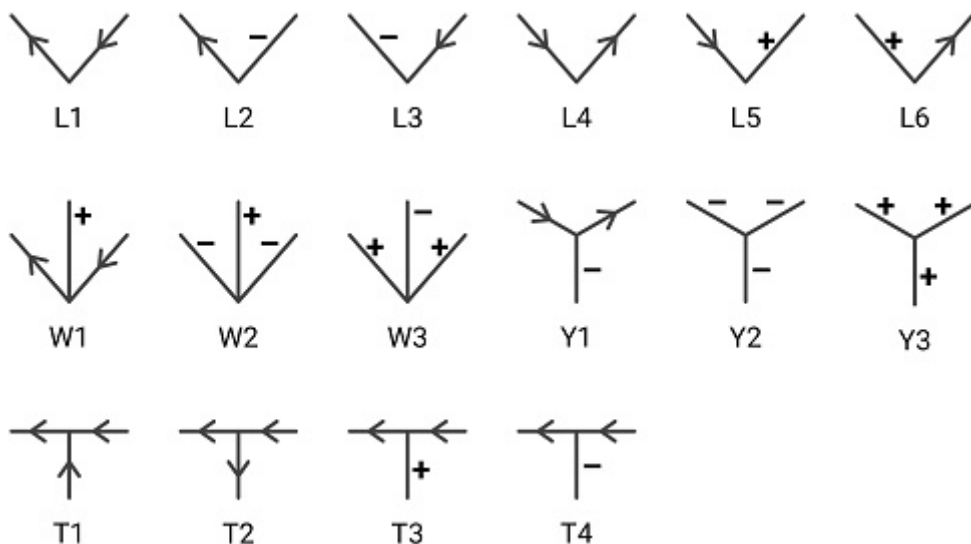
Question Label : Comprehension

CONSTRAINT SATISFACTION

The set of junctions (L, W, Y and T type junctions) that occur in a 2D line drawing of trihedral objects is provided below. The in-plane clockwise/counterclockwise rotations of these junctions are valid as well. These junctions provide constraints on the possible edge assignments (convex, concave, arrow) for the edges/lines in 2D line drawings of trihedral objects.

The junctions carry unique labels: L1, L2, L3, L4, L5, L6, T1, T2, T3, T4, W1, W2, W3, Y1, Y2, Y3. When

required, use the labels in short answers.



Note: A 2D line drawing of trihedral objects is considered to be consistent if all the edges and junctions can be assigned labels that are consistent with each other, otherwise the drawing is considered to be inconsistent and all labels are reset to NIL.

Apply a suitable algorithm to assign consistent labels to edges/junctions in the 2D line drawings in the sub-questions. Choose a suitable edge and junction order for solving the problems.

Based on the above data, answer the given subquestions.

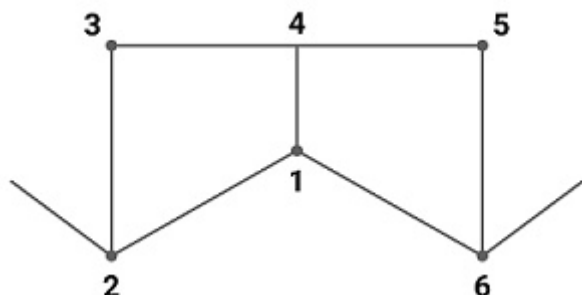
Sub questions

Question Number : 99 Question Id : 6406531250359 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Assign consistent labels to all the edges and junctions in the 2D line drawing shown below. Enter the labels of the junctions 1, 2, 3, 4 in the text box, in that order. Or enter NIL if the drawing has no consistent label assignment.



Enter a comma separated list of junction labels, or enter NIL.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X9,Y9,Z9,W9

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes